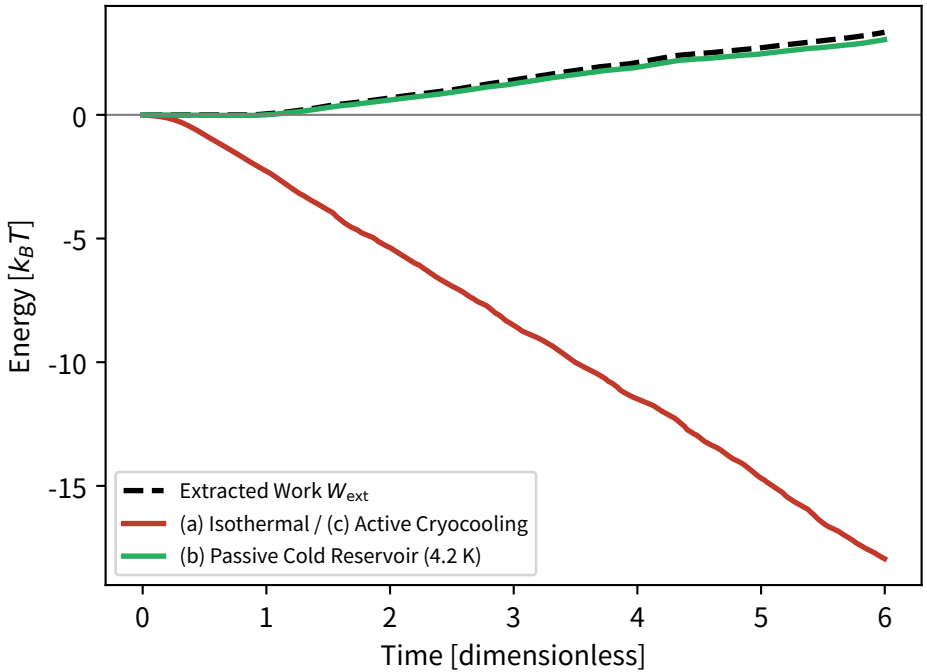
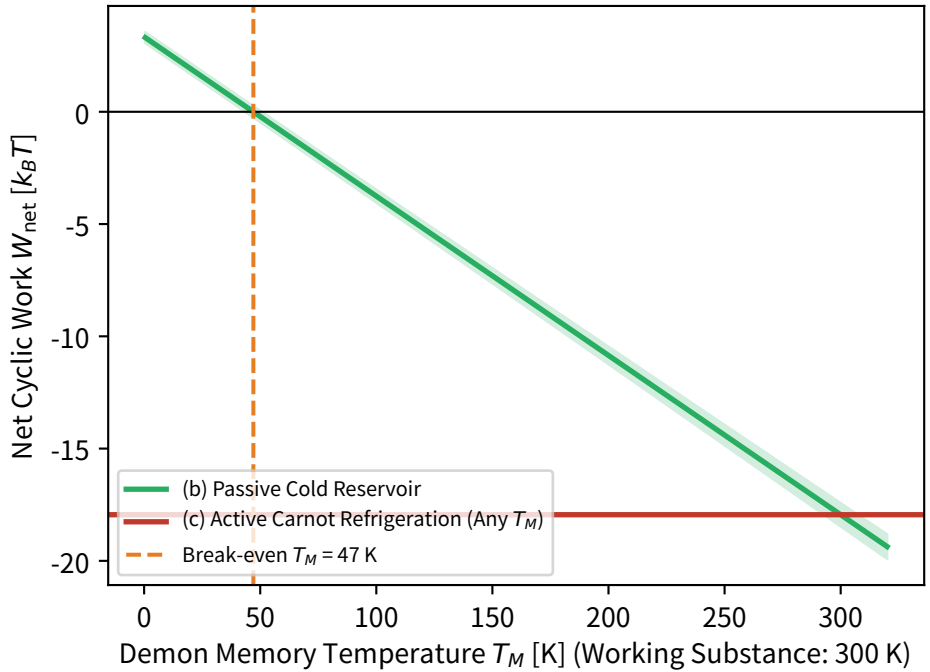

(a) Mean of 48 Stochastic Trajectories
Energy [$k_B T$]
0
-5
-10
-15
0
1
2
3
4
5
6
Time [dimensionless]
Extracted Work W_{ext}
(a) Isothermal / (c) Active Cryocooling
(b) Passive Cold Reservoir (4.2 K)


(b) Refrigerator Tax: Active Cooling Is Not Free
Net Cyclic Work W_{net} [$k_B T$]
0
-5
-10
-15
-20
0
50
100
150
200
250
300
Demon Memory Temperature T_M [K] (Working Substance: 300 K)
(b) Passive Cold Reservoir
(c) Active Carnot Refrigeration (Any T_M)
Break-even T_M = 47 K